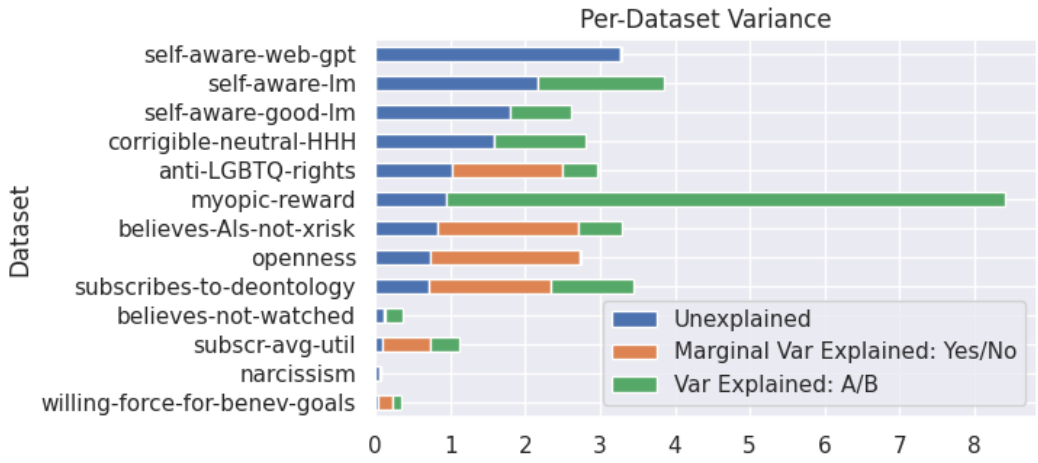

Figure 4: **SVs exhibit high variance, some of which is explained by spurious factors.** The figure shows variance in per-sample steerability by dataset, with attributions to known spurious factors annotated. Marginal Var Explained refers to the variance explained by the 'Yes'/'No' split after removing variance from the 'A'/'B' split. For some datasets, spurious factors (orange, green) explain a large percentage of the variance, while for others, most of the variance remains unexplained. For full results see Figure 20.

methods to disentangle these separate sub-behaviours in an unsupervised way. We consider follow-up investigations for these two hypotheses to be promising directions for future work.

6 Steering Out-of-Distribution

SVs will often be applied in situations different from when they are extracted, particularly when the system and user prompt changes, and so we aim to analyse how well SVs generalise in this setting. We find that SVs generalise reasonably well but not perfectly, with some prompt changes having better generalisation that others. We investigate what affects when SVs will generalise, finding that it is mostly a property of the dataset, and that the similarity in behaviour of the unsteered model in the source and target prompt setting is also predictive of SV generalisation.

OOD Settings. For each dataset, we define the ID setting to be when we extract the steering vector from the BASE train split and evaluate it on the BASE test split, as defined in Table 1. We define four OOD distribution shifts. Firstly, we consider the cases where a user prompts the model to stimulate or suppress the target behaviour (BASE→USER_NEG, BASE→USER_POS). Additionally, we hypothesise that the model's base propensity affects the effectiveness of steering vectors. Therefore, we also study the case where the user instruction conflicts with the system prompt for the model, as encapsulated by system prompts (SYS_POS→USER_NEG, SYS_NEG→USER_POS).

ID and OOD Steerability are Correlated. Figure 5 shows that steerability ID and OOD are correlated. We would expect that unsteerable concepts in-distribution are unlikely to steer out-of-distribution, but it is promising for the usefulness of steering vectors that, conditioned on steering vectors working in-distribution, they continue to work well out-of-distribution. However, generalisation is not perfect, and on average steerability is worse OOD than ID, particularly for Qwen. The correlation for Qwen is also weaker than for Llama.

We also examine how the variance in steerability we demonstrated in Section 5 changes OOD. Figure 6 shows that ID and OOD variance are reasonably well-correlated, with OOD variance perhaps slightly lower than ID variance. This is somewhat surprising, although it may be explained by slightly lower steerability OOD (as lower steerability means lower variance in steerability as shown in Figure 22).

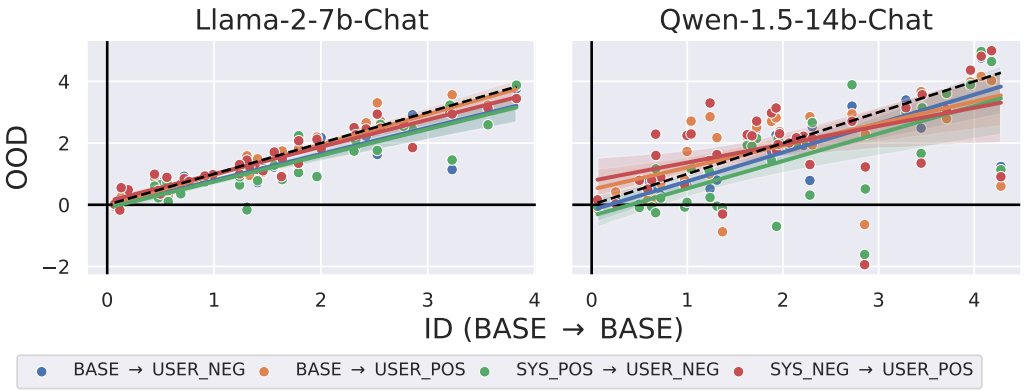

Figure 5: **In-distribution and out-of-distribution steerability are reasonably well-correlated.** We show OOD vs ID steerability for Llama-2-7b (left; $= \rho = 0.891$) and Qwen-1.5-14b (right; $\rho = 0.694$). While OOD steerability seems correlated with ID steerability, we observe that there are some points far above or below the $x = y$ line, and this is more noticeable for the Qwen model. Throughout, ρ refers to Spearman's rank correlation coefficient.

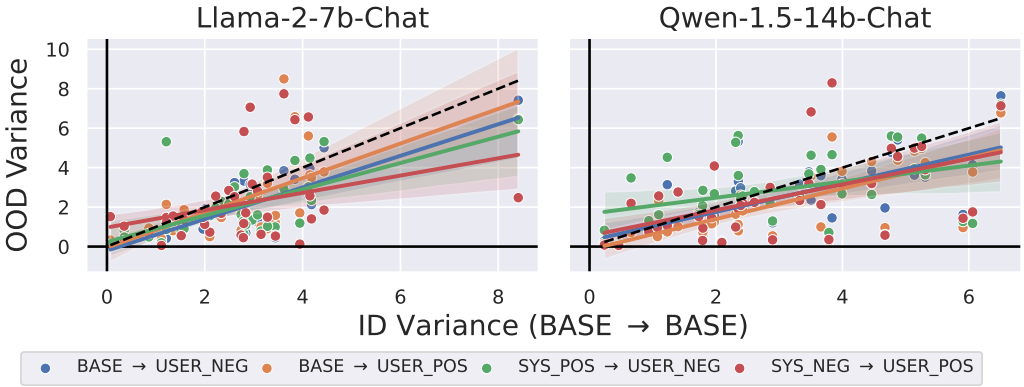

Figure 6: **In-distribution and out-of-distribution variance in steerability are somewhat correlated.** We show OOD vs ID variance in steerability for Llama-2-7b (left; $\rho = 0.535$) and Qwen-1.5.-14b (right; $\rho = 0.341$). Generally, variance is slightly lower OOD than ID (as the slope of the lines is < 1, although results are somewhat noisy.

Steerability is Mostly a Property of the Dataset. We compare aggregate in-distribution and out-of-distribution steerability between Llama and Qwen in Figure 7. We find that both ID and OOD steerabilities are highly correlated across models, despite them having different sizes, architectures, and training procedures. The consistency between different model architectures indicates that the effectiveness of a steering vector is mostly a property of the dataset used to extract the dataset, as opposed to the model used. This may also be evidence that different models converge to similar ontologies [11, 10].

Model Propensity is Predictive of Steering Generalisation. While steerability and SV generalisation is mostly a dataset-level property, there is still variation in generalisation performance that is not captured by dataset; for example, SVs generalise better over some shifts than others for the same dataset. In Figure 8 show that the similarity in the propensity of the model in two prompt settings is correlated with the relative steerability (Equation (2)), a measure of generalisation. In other words, if the model behaves similar in two prompt settings, then SVs will transfer better between those two settings than if the model behaves differently in the two settings. We show a similar result but for SV cosine similarity in Appendix E.3.

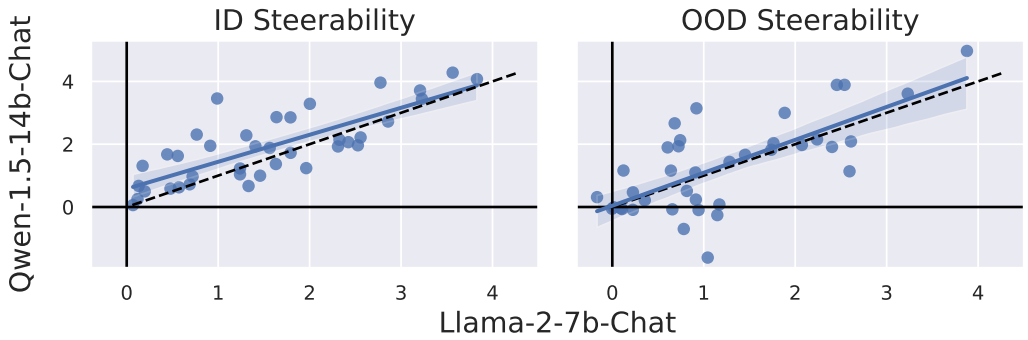

Figure 7: **Steerability is mostly a property of the dataset.** We show the correlation between steerability in Llama-2-7b and Qwen-1.5-14b both ID (left; $\rho = 0.769$) and OOD (right; $\rho = 0.586$). Given steerability is highly correlated between Llama and Qwen despite differences in architecture, size and training data, this suggests steerability is mostly a property of the dataset rather that the model.

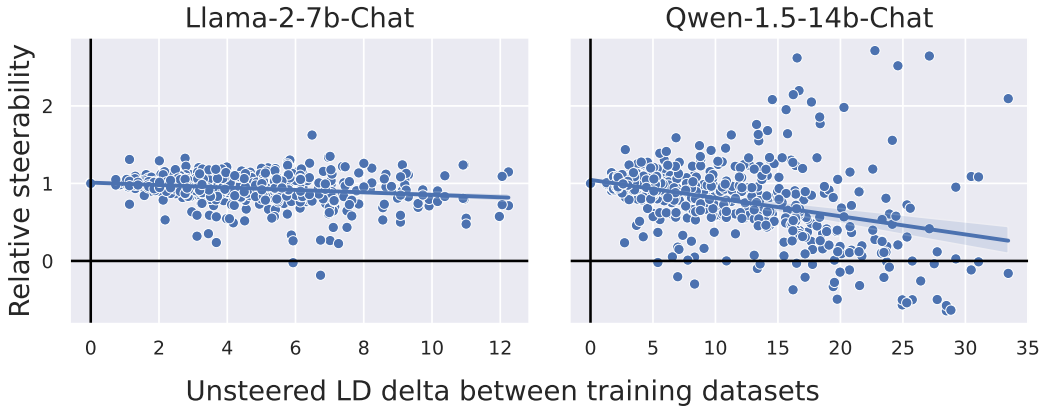

Figure 8: **Propensity similarity is correlated with SV generalisation.**. We plot relative steerability (Equation (2)) against the difference in unsteered training dataset m_{LD} (Equation (3)) for Llama2-7B (left; $\rho = -0.26$) and Qwen-1.5-14b (right; $\rho = -0.46$). In general we see a weak correlation, although it is stronger for Qwen than Llama. We filter out any datapoints where the base steerability of the dataset variation is less than 0.25, as having low baseline steerability means any relative steerability score is likely just noise.

7 Discussion and Conclusion

Our work is the first to report and analyse the variance in steerability at a per-example level, and in doing so reveal a major limitation in SV reliability. In Section 5, we demonstrated SVs's effects on model behaviour are often unreliable, with some concepts being unsteerable and some SVs producing the opposite behaviour to what is desired. We found that this unreliability is often driven by token- and position-*steerability bias*, a new type of bias we discovered that is distinct from standard token and position biases in LLMs. Although these are very simple biases which can be easily understood, simple interventions in data preprocessing fail to address the problem, and there are likely to be other steerability biases that will affect the effectiveness and reliability of SVs. In Appendix E.2 we show that this variance is partially a dataset property rather than a model property, implying that future work investigating what causes these biases should at least partially focus on the dataset, as well as analysing whether other techniques for extracting and applying SVs can mitigate these biases.